



LINKED LISTS VS ARRAY

Software Design 3

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Abstract

The aim of this document is to give an overview on how the assignment of memory management is tackled. Therefor there will be discussed on the research that has been done, and the solution which will be implemented.

Table of Contents

1. Introduction	2
2. Research	2
2.1. Linked List API	2
3. Design	3
3.1. freeMemory	3
3.2. claimMemory	3
4. Reflection	4
5. References	4

Table of figures

Figure 1. Flowchart freeMemory	3
Figure 2. Flowchart claimMemory	3

1. Introduction

When programming, there is made use of different kind of memory within the program. Each kind of memory serves their own purpose. For example the global variables are permanent storage, local variable are stack and free memory is the heap. The heap is a flexible memory area and is influenced by allocating memory while a program is being executed.

To get a better understanding of how the heap exactly works, therefor is this assignment. By creating two linked lists, one which simulates the free memory and the other which simulates the allocated memory, there can be displayed how they work together and how they behave in general.

Regarding to the exact requirements for this assignment, see (Frenken, Assignment 2: Linked Lists vs Arrays, 2021).

2. Research

To succeed with the given assignment, there were multiple things to be researched beforehand and during programming the program itself. Beforehand there have been created a flowchart for claiming memory and freeing memory, see 3. Design. For the other researched topics, see the subheadings of 2. Research.

2.1. Linked List API

For this assignment there is the possibility to reuse parts from the API, that was created during the part 1 of (Frenken, Assignment 2: Linked Lists vs Arrays, 2021). This is due to the fact that there is made use of adding and removing nodes in a list and since the used lists are linked as well. Therefor whenever there is made a call to , for example, claim memory there will be a 'block' removed from the freeList and added to the allocList.

3. Design

3.1. freeMemory

When memory needs to be freed there are multiple steps that need to be taken. For example the allocList needs to be manipulated in such way that the memory block is erased, and freed. Besides that the freed memory should be made available again, and therefor needs to be added to the freeList. See Figure 1. Flowchart freeMemory.

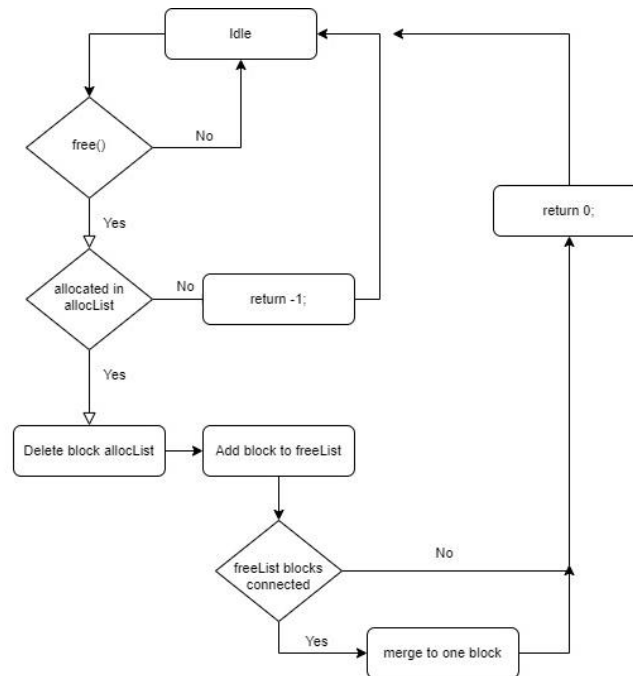


Figure 1. Flowchart freeMemory

3.2. claimMemory

When memory needs to be claimed, the process exists out of multiple steps. These steps can be divided into adjusting the blocks in the freeList and the allocating the memory in the allocList by adding a new block. See Figure 2. Flowchart claimMemory.

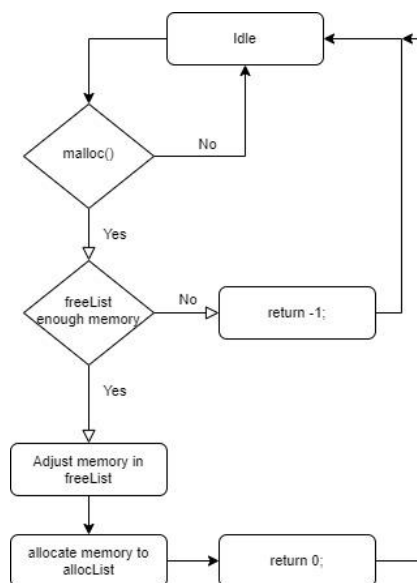


Figure 2. Flowchart claimMemory

4. Reflection

Throughout this assignment I've been able to understand the need of creating your own API's. These API's can help a lot with speeding up programming since you can implement them into your code if a similar situation occurs. Besides that, with the help of this assignment I've put on more knowledge on how malloc and free work exactly. It has given me a more clear insight.

To take a look at how the assignment went overall, overall it went well. I was able to write a code fairly quick and whenever I ran into an error, I was able to solve it with the help of debugging and (Python Tutor, n.d.). Lastly, once again I realised that I should have started earlier with the assignment.

5. References

Frenken, R. (2021). *Assignment 2: Linked Lists vs Arrays*. From git.fhict.nl: https://git.fhict.nl/technology/t-sem3-cb/-/blob/master/sd/2_Linked_Lists_vs_Arrays/Assignment_Linked_Lists_vs_Arrays.docx

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